

An overview of systematic reviews to determine the impact of socio-environmental factors on health outcomes of people with disabilities

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Abstract

People with disabilities are often subject to intersecting layers of social and economic disadvantage and other barriers that drive health inequity. As a result, they frequently experience worse health than people without disabilities, beyond the direct effects of their health condition or impairment. The aim of this overview of systematic reviews was to summarise the evidence on the impact of socio-environmental factors (i.e. social, physical or attitudinal) on the health outcomes of disabled people. A systematic search of five databases (MEDLINE, PsycINFO, Embase, CINAHL and Scopus) for English-language articles from January 2000 to April 2021 was conducted. Abstracts were screened by two reviewers and reviews were critically appraised. Key data were extracted by topic, population, disability type, critical appraisal method, socio-environmental themes and health outcomes. There were 23 systematic reviews identified examining adult (60.9%) or child and young (8.7%) disabled people, with 30.4% not specifying an age range. Reviews examined people with neurological or physical (39.1%), intellectual (17.4%), sensory (8.7%) or a range of (34.8%) disabilities. Three key health outcomes (i.e. access to healthcare, health-promoting behaviour and care quality) and several recurring socio-environmental themes related to the health outcomes of disabled people were identified. Disabled people encounter common social, physical and attitudinal factors that hinder their health outcomes in terms of access to services and quality healthcare. Many preventive health services were identified as either inaccessible or not meeting the needs of disabled people. Greater involvement of disabled people in service design and awareness raising is essential.

KEYWORDS

attitudinal, environment, health outcome, physical, social

1 | INTRODUCTION

Around 15% of the population worldwide, an estimated 1 billion people, are living with some form of disability (World Health Organization, 2011). A disability can either be permanent or

temporary, with the number of people living with a disability growing as the population ages and the prevalence of chronic health conditions increase (United Nations Department of Economic & Social Affairs: Population Division, 2020; World Health Organization, 2011). The experience of disability does not take any one shape; it is diverse,

complex and multidimensional (World Health Organization, 2011), and a disability can have an impact on a person's needs across the health, social and built environment domains.

Difficulties with service provision and accessibility, discriminatory attitudes, socioeconomic disadvantage and social exclusion are widely accepted barriers experienced by disabled people that undercut their full participation in society on an equal basis to others (Drainoni et al., 2006; Mithen et al., 2015; World Health Organization, 2011). Disabled people have reduced opportunities for education and work, greater housing insecurity, lower education and higher rates of socioeconomic disadvantage than people without disabilities (Kavanagh, 2020; National People with Disabilities & Carer Council, 2009). As a result, disabled people, many with unique health needs, often experience worse health outcomes than non-disabled people (Drainoni et al., 2006; Reichard et al., 2011).

Healthcare services may be underutilised by disabled people for a number of reasons. Common barriers to healthcare access and provision experienced by people with disabilities are physical barriers due to the inaccessibility of buildings and unadaptable medical technology and equipment (da Rosa et al., 2020), poor or unavailable healthcare services to suit their needs (Drainoni et al., 2006) and healthcare affordability (Merten et al., 2015). Moreover, healthcare staff and other service providers may have inadequate knowledge and training in caring and working with disabled people (Bowen et al., 2020). Negative social attitudes towards disability can influence healthcare provision and also peer and family relationships with a disabled person (Iacono et al., 2014; Lindsay, 2016). Disabled people have reported experiencing a lack of knowledge among health professionals of medical issues related to disability, poor healthcare coordination and insensitive interactions (Drainoni et al., 2006).

Many systematic reviews have been conducted that have examined different aspects of service accessibility and social inclusion and their impact on health outcomes of disabled people. These reviews have largely focused on accessibility issues (such as in the home or social environments) (Calder et al., 2018; Cho et al., 2016), access and use of clinical preventive services (such as health screening) (Andresen et al., 2013), poor experiences of healthcare or unmet care needs (Gibson & O'Connor, 2010; Iacono et al., 2014) and barriers to service access or participation (Barr et al., 2018; Breckenridge et al., 2014; Edwards et al., 2020; Matin et al., 2021). There has also been some exploration of perceived negative social attitudes of disability and associated experiences of people with disability in accessing healthcare (Lawler et al., 2013; Matin et al., 2021). While each review targeted a specific area relevant to the health and service needs of disabled people, the diversity and multidimensional impact on the health outcomes of disabled people are lost when each review is considered as a sole entity. The overall objective of this research is to identify and summarise the evidence on the impact of social, physical or attitudinal environmental factors on key health outcomes for disabled people. This research aims to provide an overview of systematic reviews to summarise the evidence on the impact of socio-environmental factors (i.e. social, physical or attitudinal) on the health outcomes of disabled people.

What is known about the topic?

- People with disability often have difficulty accessing preventive health services.
- Some health professionals have inadequate knowledge regarding disability-focused care.

What this paper adds?

- Common socio-environmental themes across different disabilities hindered access to services and quality healthcare.
- Essential to have involvement of people with disability in service design, messaging and awareness raising.
- Adoption of universal design principles needed for the built environment and medical equipment.

2 | METHOD

This is an overview of systematic reviews on the impact of socio-environmental factors on the health outcomes of disabled people. The overview identified systematic reviews that examined social, physical or attitudinal factors influencing health outcomes of disabled people and extracted data on the main socio-environmental themes and health outcomes were examined in each review (Pollock et al., 2021).

2.1 | Concepts and definitions

For this review, the guiding definition of disability was that it is the product of interactions between people with impairments and contextual factors, such as attitudinal or environmental barriers, that limit an individual's full and effective participation in society on an equal basis with others (United Nations, 2006; World Health Organization, 2011, 2021). Impairments may be categorised as physical, psychological, intellectual or sensory. Disability was not conceptualised as an attribute of particular people, however, the term is used throughout this review to describe the population of interest as a coherent group. For the purposes of this review, transitory forms of disability resulting from injury or illness, where the person experiencing impairment is likely to reoccupy a non-disabled social role and is expected to regain functionality, were excluded.

The Craig Hospital Inventory of Environmental Factors (CHIEF) (Craig Hospital Research Department, 2001) was used as a framework to conceptualise socio-environmental factors that may impact on the experiences of disabled people. The constructs in CHIEF have strong parallels and overlap with the socio-environmental factor domains of the International Classification of Functioning, Disability and Health (World Health Organization, 2021) and the Centres for Disease Control and Prevention's Disability Barriers to Inclusion (Centers for Disease Control & Prevention, 2020), however, the

CHIEF is able to encompass the breadth of socio-environmental factors. The CHIEF factors include the following: policies (e.g. employment policies, community service policies), physical or structural (e.g. design of community, technology, natural environment), work or school (e.g. support and attitudes at work and school), attitude or support (e.g. discrimination, support and attitudes in the home) and services or assistance (e.g. transportation, education and training, help in the community).

Health outcomes included those relating to healthcare utilisation and care quality, physical and mental health, health-related quality of life (HRQoL), ability to perform activities of daily living (ADLs), and participation in health-promoting behaviours such as physical activity. For the purposes of this overview, outcomes more distally related to health, including social determinants of health (such as education, employment or housing) were excluded.

2.2 | Data sources and eligibility criteria

A systematic search was conducted using five databases: MEDLINE, PsycINFO, Embase, CINAHL and Scopus. The search strategy was developed with a university librarian to include search terms and keywords related to three key concepts: (i) disability; (ii) environmental factors and (iii) health outcomes. The base keywords for the search were as follows: (i) disabled persons OR amputee OR persons with mental disability; (ii) environment OR policy OR workplace OR attitude OR social inclusion and (iii) health services OR accessibility OR healthcare accessibility OR outcome. The exact search string was adapted to optimise the search of each database (see Appendix S1 for full search strategy). Results were limited to English-language articles that were published in peer-review journals from 1 January 2000 to 7 April 2021. Snowballing of references included in the reviews was conducted to identify any potential reviews not previously identified. The BMC Women's Health and the Disability and Health Journal were hand-searched by one reviewer (TR) for any additional reviews.

2.3 | Review screening

The review protocol was registered with PROSPERO International Prospective Register of Systematic Reviews (ID:CRD42021253149) and conducted using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) method. Systematic reviews were included if they examined the impact of socio-environmental factor(s) on the health outcome(s) of disabled people. Reviews of studies from solely low- or middle-income countries were excluded due to general differences in access to healthcare and services compared to high-income countries (Hashemi et al., 2020; World Health Organization, 2011). Reviews were excluded if disabled people were not the primary population of interest, if insufficient detail was given about the socio-environmental factor(s) or health outcome(s), or if the review primarily related to clinical, behavioural, surgical or

pharmacological interventions without also locating disability within a broader social environmental context (i.e. only considered within a biomedical lens) (Wade & Halligan, 2004). Whilst recognising that ongoing disability can be acquired through traumatic injury, articles which focused on the acute care or adjustment periods following injury were excluded.

The results of the database searches and journal hand-search were imported to Endnote X9 and duplicates removed. The abstracts were independently assessed for inclusion by two reviewers (TR, KM), who met regularly to discuss uncertainties. If it was clear from the abstract that the article was not a systematic review or did not examine the impact of socio-environmental factors on the health outcomes of disabled people it was excluded. Both reviewers (TR, KM) screened 10% of the articles and interrater reliability was >94% agreement. Any disagreement on abstracts were discussed and 100% consensus achieved.

2.4 | Data extraction and quality review

If an article was not excluded in the abstract review stage, the full text of each included review was assessed by two reviewers (TR, KM) and then appraised by a third reviewer (RM). Where available, key information extracted included: authors and publication year; review objective/aim; topic, characteristics of included studies, population characteristics, type of disability, socio-environmental factors, health outcomes, critical appraisal tool used and summary findings. The methodological quality of reviews was assessed by two reviewers (TR, KM) using the Joanna Briggs Institute's (JBI's) critical appraisal checklist for systematic reviews and research syntheses (The Joanna Briggs Institute, 2017). Reviews were rated as low- (<5 items), mid- (6–8 items) or high- (>9 items) quality according to the number of checklist items fulfilled. Reviews were not excluded on the basis of quality appraisal. However, the low-quality reviews did not contribute evidence that was not elsewhere supported by a review with a higher quality review.

2.5 | Data synthesis

The key characteristics of included reviews were extracted and tabulated. The socio-environmental factors examined and described in the results and discussion sections of included reviews that were indicated to influence and/or have an impact on health outcomes of people with disabilities were categorised based on CHIEF themes (Craig Hospital Research Department, 2001) by two reviewers (TR, KM) and then appraised by a third reviewer (RM). Two reviewers (TR, KM) undertook an inductive, iterative process to categorise the socio-environmental factors using the CHIEF (Craig Hospital Research Department, 2001). Influencing factors were initially identified in each review article (e.g. absence of personal assistive devices) and then classified into broad high-level themes (e.g. services—personal equipment) to describe the breadth

of socio-environmental factors identified for the health-related outcomes examined (Table 1). Multiple health outcomes and socio-environmental themes and sub-themes could be identified in each review.

3 | RESULTS

There were 614 potentially eligible systematic review articles identified after removing duplicates. Most articles were excluded at the abstract screening stage ($n = 580$; 94.5%). Thirty-four full-text articles were assessed and, of these, 11 (32.3%) were excluded after full-text review. There were 23 reviews that met all study criteria and were included in the synthesis (Figure 1).

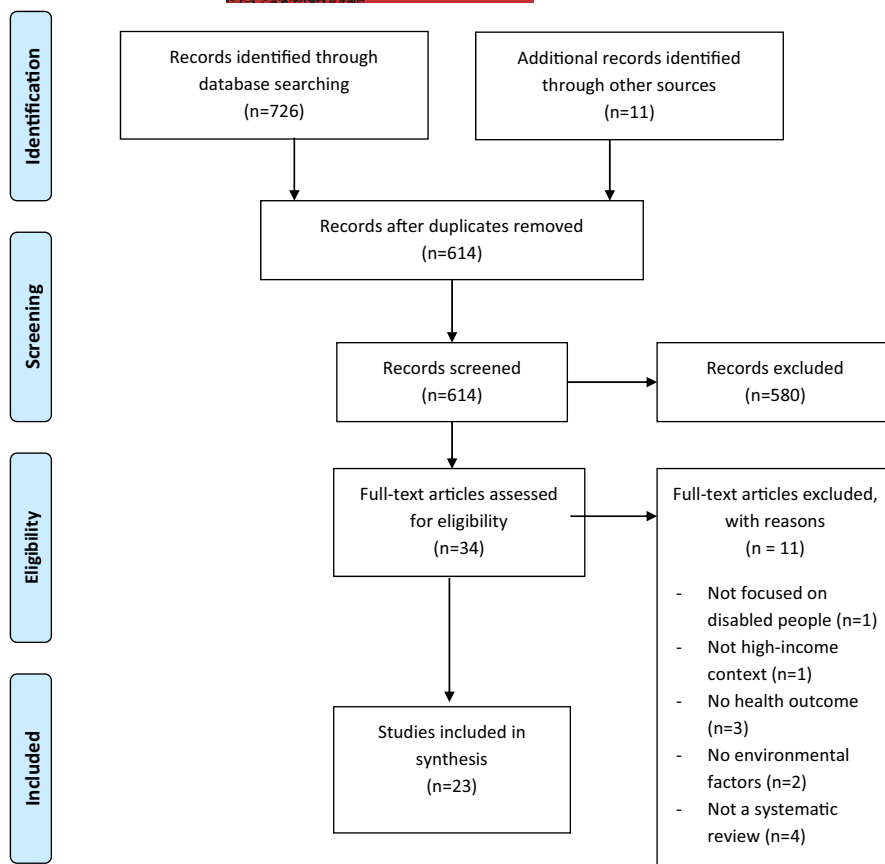
Of the 23 included reviews, 14 (60.9%) were focused exclusively on adult populations, two (8.7%) reviews looked exclusively at children and/or young people and seven (30.4%) did not specify an age range. Seven (30.4%) reviews specifically focused on women with disabilities. In terms of type of disability, nine (39.1%) reviews focused on people with neurological or physical disabilities, four (17.4%) exclusively looked at learning, intellectual or developmental disabilities, two (8.7%) solely focused on people with sensory disabilities (i.e. visual or hearing impairment) and eight (34.8%) reviews did not specify the type(s) of disability examined. The included reviews examined a range of health outcomes, including access and utilisation of health services ($n = 13$) (including oral health, audiology, oncology and maternal and infant health), participation in health behaviours ($n = 8$) (including diabetes self-care, menstrual hygiene management

TABLE 1 Environmental factor theme conceptual definitions

CHIEF theme	Sub-theme	Conceptual definition
Policies	Policy—government	Refers to government policies and guidelines for service provision, government funding for social services and healthcare. Includes accessibility legislation.
	Policy—financial	Refers to private health insurance policies and eligibility for coverage, out-of-pocket expenses for healthcare, cost of participating in programs.
	Policy—organisation	Refers to the policies and rules of organisations and businesses, e.g. appointment scheduling, eligibility criteria for services, training requirements for staff.
Physical or structural	Physical—environment	Refers to the natural and built environment, particularly relating to public spaces and amenities, e.g. footpaths, parking spaces, building entryways.
	Physical—facility/equipment	Refers to the design and accessibility or adaptability of particular facilities and machines, devices, technology within facilities, e.g. exercise equipment, scanning machines.
Services or assistance	Services—available programs/services	Refers to the absence or presence of programs (e.g. parenting courses, fitness classes) or services (e.g. allied healthcare, audiology). Also includes the availability of trained staff and specialists in the local area.
	Services—information provision/communication	Refers to the availability and provision of information in an accessible format, and interpersonal communication that accommodates differences in how people receive messages and express themselves. Also includes language style, i.e. sentence complexity, jargon and technical terms.
	Services—transport	Refers to availability, affordability, accessibility of public or private transportation. Including the routes and distance of transportation stops (e.g. bus stops, train stations) to services.
	Services—personal equipment	Refers to personal equipment, products or devices, such as adapted insulin monitoring devices and sports wheelchairs. Includes the affordability and acceptability of the product.
Attitude or support	Attitudes and support—providers, community, family	Refers to the attitudes, support and encouragement given to people with disabilities from family and peers, the community, service providers and healthcare professionals. Also includes stereotyping, stigma and discrimination.
	Knowledge/training—providers	Refers to the knowledge and awareness held by service providers and professionals, including healthcare workers, of how to engage appropriately with people with disabilities. The availability of evidence to inform care for people with disabilities. Training in working and communicating with people with disabilities (and carers).

Abbreviation: CHIEF, Craig Hospital Inventory of Environmental Factors.

FIGURE 1 PRISMA flow diagram



and physical activity) and the experience and quality of care in acute care settings ($n = 6$) (Table 2). Most reviews were assessed to be of high- ($n = 16$; 69.6%) or mid- ($n = 6$; 26.1%) quality. Only six (26.1%) reviews assessed the likelihood of publication bias (Table 3). Illustrative quotations from reviews for socio-environmental themes and sub-themes are shown in Table 4.

The two most frequent reoccurring socio-environmental themes identified that influenced or had an impact on access to healthcare of disabled people were knowledge and training of providers ($n = 9$) and the physical design of facilities and equipment ($n = 9$). Care quality of disabled people was most frequently influenced by information provision and communication ($n = 4$) and health-promoting behaviours of disabled people was frequently influenced by available programs ($n = 5$), knowledge and training of providers ($n = 4$), the physical design of facilities and equipment ($n = 4$) and the transportation services ($n = 4$) socio-environmental themes (Figure 2).

4 | DISCUSSION

This overview of systematic reviews sought to collate and summarise the evidence on the impact of social, physical or attitudinal environmental factors on a number of key health outcomes for disabled people. It identified several common recurring CHIEF socio-environmental themes and sub-theme from the reviews that were identified as having an impact on health outcomes (i.e. access to healthcare, health-promoting behaviours and care quality) of disabled people.

4.1 | Knowledge, training and attitudes of providers, family and carers

Several reviews described the poor experience and quality of healthcare received by disabled people in clinical settings, largely attributed to care providers' lack of training in providing disability-focused care and unwillingness or confidence to adapt existing models of care (Bradbury-Jones et al., 2013; Breckenridge et al., 2014; da Rosa et al., 2020; Edwards et al., 2020; Gibson & O'Connor, 2010; Heideveld-Gerritsen et al., 2021; Lawler et al., 2013; Marrocco & Krouse, 2017; Matin et al., 2021). In some situations, healthcare providers may feel anxious and ill-equipped to provide care to disabled people with complex care needs who may also have communication barriers, thus contributing to poor experiences of healthcare by disabled people (Bradbury-Jones et al., 2013; Marrocco & Krouse, 2017). Moreover, it appears there is a lack of evidence-based care guidelines for people with disabilities (Homeyard et al., 2016). As a consequence of the lack of evidence, practitioners rely on protocols devised for people without disability, such as protocols for maternal care for women (Homeyard et al., 2016; Lawler et al., 2013; Matin et al., 2021). Ensuring that care protocols include information on how care plans can be adapted for people with a range of disabilities may go some way to providing healthcare providers with information on how best to deliver disability-focused care.

At the same time, the beliefs of healthcare professionals, educators and the wider community are influenced by prevailing

social attitudes that often devalue people with disabilities. Several reviews drew attention to the experiences of disabled people (women in particular) feeling unsafe, being left unattended and feeling disrespected in healthcare settings, and the perception of health professionals as unwilling to accommodate diversity (Lawler et al., 2013; Matin et al., 2021). Concerningly, two reviews highlighted that healthcare and other professionals disregarded the preferences and input of disabled people and their carers (Bradbury-Jones et al., 2013; Horner-Johnson et al., 2019). Providers' misconceptions about the needs and priorities of disabled people have also determined whether disabled individuals were referred to, or gained access to, services and treatments such as cancer screening and contraceptives (Breckenridge et al., 2014; Edwards et al., 2020; Horner-Johnson et al., 2019). Some care tasks, such as menstrual care, were identified as being disliked by professional carers, with limited support also provided to families and the person with a disability on how to manage menstrual hygiene (Wilbur et al., 2019). It appears there are likely to be opportunities to redress social attitudes, perceptions and training of care providers in how to work and care for people with disabilities, such as through induction training or ongoing professional development (Pelleboer-Gunnink et al., 2017; Satchidanand et al., 2012).

Three reviews acknowledged the importance of considering multiple marginalisation due to the intersectionality between disability and other minority identities (Barr et al., 2018; Lawler et al., 2013; Trounson et al., 2020). Societal marginal groups, such as stemming from cultural or linguistic differences, can further have an impact on health outcomes for disabled people. In addition to disability-related barriers, there are both language and cultural barriers to accessing healthcare for Indigenous people (Barr et al., 2018), people in minority ethnic groups (Robertson et al., 2019), and among lesbian, gay, bisexual, transgender, intersex or questioning people with a disability (Wilson et al., 2018). Greater involvement of consumers with a disability, including those in further marginalised groups, in service design and providing cultural competency awareness for healthcare providers (Olaussen & Renzaho, 2016; Trounson et al., 2020) are two potential strategies to reduce service and care inequalities (Beverley et al., 2004; Trounson et al., 2020).

The role that carers and supporters play in assisting people with disabilities, accommodating their role within healthcare spaces and providing practical support was acknowledged in the reviews. However, two support mechanisms of carer training and the importance of the availability of respite care were viewed as additional strategies that could positively impact the health and wellbeing of disabled people (Bradbury-Jones et al., 2013; Wilbur et al., 2019; Yang et al., 2020). Two reviews flagged that opportunities for increasing awareness among family members and carers of the need to provide support to the disabled person in overcoming barriers (e.g. transportation, informational) to service access and participation were required (Bloemen et al., 2015; Bodde & Seo, 2009).

4.2 | Physical environment, facilities, equipment

The physical environment can pose barriers to accessing the space where health services are provided. People with neurological and physical disabilities were commonly the focus of reviews which considered the impact of the built environment on the use of health services, especially preventive health services and screening (Edwards et al., 2020; Lawler et al., 2013; Marrocco & Krouse, 2017). Despite the existence of universal design principles, examples of inaccessibility in the built environment included narrow doorways, doors without a push button for automatic opening, impaired access to toilet facilities, lack of ramps at entranceways and a lack of disabled parking spaces (Breckenridge et al., 2014; da Rosa et al., 2020; Edwards et al., 2020; Heideveld-Gerritsen et al., 2021; Lawler et al., 2013). Barriers in the built environment were common across facilities (Bloemen et al., 2015; Eisenberg et al., 2017). At recreational facilities, for instance there may be no place to leave a wheelchair close to exercise equipment, or no access seat or hoist (or staff sufficiently trained) to enter a swimming pool (Martin Ginis et al., 2016; Newitt et al., 2016). Such barriers can discourage, if not make entirely unavailable, the use of facilities for physical activity by disabled people. In public spaces, design features such as footpaths, lighting and street density may encourage leisure-time physical activity and social participation (Eisenberg et al., 2017). Ensuring that universal design principles are adopted and accessibility issues are considered and addressed when designing and building healthcare facilities will enable these facilities to be physically accessible by all individuals.

The absence of adjustable equipment and devices can make it difficult for disabled people to undergo preventive health screening and examinations safely and comfortably. For instance three reviews noted that mammography machines are typically designed to be used while a person is standing upright and may not be adjustable to a lower height (Edwards et al., 2020; Gibson & O'Connor, 2010; Marrocco & Krouse, 2017). These obstacles may render services totally inaccessible, unable to be accessed without assistance or prior planning, or may be so burdensome and undignifying to use that disabled people are excluded or deterred from their utilisation (Marrocco & Krouse, 2017; Matin et al., 2021). In fact, it has been estimated that disabled people only receive around 35%–40% of recommended preventive health services (Marrocco & Krouse, 2017). Equipment that adapts processes to suit the needs of disabled people, such as an adaptive wheelchair that enables participation in sport, can facilitate participation by disabled people (Bloemen et al., 2015). However, there can be difficulties (such as cost) associated with obtaining adaptive equipment or assistive devices for personal use (Bloemen et al., 2015; Mulligan et al., 2012; Newitt et al., 2016).

4.3 | Transport and service or program availability

Across the reviews, and regardless of impairment, transportation was commonly identified as a key factor influencing a disabled

TABLE 2 Characteristics of the systematic reviews examining the impact of environmental factors on health outcomes of disabled people

Authors	Review aim	Studies	Population	Disability
Barr et al. (2018)	Evaluate literature on service provision to children with hearing loss living in regional, rural, remote Australia	28 studies: 24 qualitative or quantitative, 3 quasi-experimental, 1 narrative review	Children ≤16 years	Sensory disability
Beverley et al. (2004)	Identify health information needs of visually impaired people and highlight major gaps in the literature	16 studies: 5 qualitative, 5 quantitative, 5 mixed method, 1 seminar	Visually impaired people, all ages	Sensory disability
Bloemen et al. (2015)	Summarise important factors associated with participation in physical activity in children and adolescents with physical disabilities	18 studies: 12 quantitative, 6 qualitative	Children and young people, >4 and ≤18 years	Neurological or physical disabilities
Bodde and Seo (2009)	Determine what environmental and social barriers prevent adults with intellectual disabilities from engaging in physical activity	7 studies: 5 qualitative, 1 quantitative, 1 mixed-method	Adults ≥18 years	Intellectual or learning disability
Bradbury-Jones et al. (2013)	Identifying influences on the health, safety and welfare of adults with learning disabilities in acute hospitals.	7 studies: all qualitative	Adults, age unspecified	Intellectual or learning disability
Breckenridge et al. (2014)	Explored how domestic abuse impacts on disabled women's access to maternity care	11 studies: 8 quantitative, 1 qualitative, 2 mixed method	Young people and adults (women ≥16 years)	Non-specific disability
Da Rosa et al. (2020)	Identify barriers (and facilitators) of access to oral health services for disabled people	16 studies: all quantitative	Non-specific	Non-specific disability
Edwards et al. (2020)	Identify the barriers (and facilitators) of access to cancer services for physically disabled adults and their experiences of cancer care	17 studies: 10 qualitative, 7 quantitative	Adults ≥18 years	Physical disability

Environmental factor themes identified	Health outcome examined	Quality assessment tool used	Author conclusions
Services (transport; information provision/communication; available programs)	Access to healthcare—specialised health services	John Hopkins Evidence-based rating scale for strength and quality of evidence Newhouse et al. (2007)	Children with hearing loss living in regional, rural and remote Australia experience reduced quality and frequency of audiology services.
Physical (environment) Services (information provision/communication)	Access to healthcare—accessibility of health information	Critical Skills Training in Appraisal for Librarians checklist Booth and Brice (2001)	Informational needs included information on health promotion, visual impairment in general, coping with visual impairment or accessing health services. Information gaps included need for diverse information and involvement of visually impaired people.
Physical (facility/equipment) Attitudes (providers and peers) Services (available programs)	Health promoting behaviour—physical activity participation	McMaster Critical Review Forms (Law et al. (1998) and Letts et al. (2007)	Factors associated with participation in physical activity included personal (e.g. physical fitness, self-efficacy, motivation, age, time) and environmental (e.g. social support, socialisation, availability of equipment, availability of local facilities).
Policy (organisation; financial) Attitudes (providers and family) Services (transport; available programs)	Health-promoting behaviour—physical activity participation	Hierarchy of evidence for assessing qualitative health research Daly et al. (2007) and Russell and Gregory (2003)	Common barriers identified were transportation issues, financial limitations, lack of awareness of options, lack of support from caregivers and authority figures (i.e. parent, teacher, coach), lack of clear policies for engaging in regular physical activity in residential and day service programs.
Policy (financial) Physical (facility/equipment) Attitudes (providers) Services (information provision/communication) Knowledge/training (providers)	Care quality—hospital setting	A global assessment of study quality Walsh and Downe (2006)	Six key areas of influence identified: care provision (meeting health and personal needs), communication, staff attitudes, staff knowledge, supporters, carers (valuing their role) and physical environment.
Policy (organisation; financial) Physical (environment; facility/equipment) Attitudes and support (providers and community) Services (transport; available programs) Knowledge/training (providers)	Access to healthcare—maternal and infant health	Critical Appraisal Skills Programme (CASP) Cohort Study Checklist Critical Appraisal Skills Programme (2018a) and a global assessment of study quality Walsh and Downe (2006)	Found that a mental health diagnosis, poor relationships with health professionals and environmental barriers (i.e. physical, geographical and institutional) can compromise use of maternity services.
Policy (financial) Physical (facility/equipment) Services (transport) Knowledge/training (providers)	Access to healthcare—oral health	Downs and Black scale for study quality Downs and Black (1998) and adapted Cochrane Collaboration tool for risk of bias Higgins et al. (2011)	Barriers identified were dentist's lack of preparation for disabled people, treatment cost, physical access barriers to dentist's facilities, communication difficulties and lack of awareness of the need for dental treatment. None of the studies discussed facilitators.
Policy (financial; organisation) Physical (facility/equipment) Attitudes (providers) Services (transport; information provision/communication) Knowledge/training (providers)	Access to healthcare—preventive, cancer screening	Mixed Methods Appraisal Tool (MMAT-Version 11) Pace et al. (2012)	Various barriers identified including not engaging in preventive health behaviour (e.g. limited knowledge of preventive healthcare, time constraints), negative interactions with health professionals (e.g. poor attitude or behaviour, lack of knowledge about disabilities), external factors (e.g. cost, transport), accessibility of facilities and equipment (e.g. physical access and parking, adaptive equipment), physical positioning concerns (e.g. need for physical support, pain and discomfort). Practical facilitators that could improve screening were disabled parking spaces, handrails by entrance, level or ramped entrances, accessibility buttons on doors, adaptive equipment in the clinic, assistance to get around buildings/undress for screening, a provider who understands disability, having additional time in the appointment.

(Continues)

TABLE 2 (Continued)

Authors	Review aim	Studies	Population	Disability
Eisenberg et al. (2017)	Whether built environment factors moderate the association between having a disability and lower levels of physical activity	15 studies: 9 quantitative, 6 qualitative	Adults ≥18 years	Physical, sensory or intellectual disability
Gibson and O'Connor (2010)	Determine if disability limits access to healthcare and identify what body functions, structures and activities and participation, as well as contextual factors (environment and personal) interact with the health condition to limit access	60 studies: 27 database studies (unspecified further), 20 quantitative, 13 qualitative	Children and adults, age unspecified	Non-specific disability
Heideveld-Gerritsen et al. (2021)	Identify and provide an overview of maternity care experiences of physically disabled women	10 studies: 9 qualitative, 1 mixed method	Adults (women, age unspecified)	Physical, including sensory disability
Homeyard et al. (2016)	Identify and review the existing evidence on the provision of antenatal care among women with an intellectual disabilities	16 studies: 9 qualitative, 5 quantitative, 2 mixed method	Adults (women, age unspecified)	Intellectual or learning disability
Horner-Johnson et al. (2019)	Identify contraceptive knowledge, attitudes and preferences regarding contraception, contraceptive use, barriers (and facilitators) to informed contraceptive use and effectiveness of interventions to improve informed contraceptive decision-making and use.	54 studies: 48 observational, 6 interventional (unspecified further)	Adults (women of reproductive age)	Non-specific disability
Lawler et al. (2013)	Identify challenges (and facilitators) to accessing maternity services experienced by women with a physical disability during pregnancy, childbirth and early motherhood	28 studies: 18 qualitative, 9 quantitative, 1 mixed method	Adults (women, age unspecified)	Physical disability

Environmental factor themes identified	Health outcome examined	Quality assessment tool used	Author conclusions
Physical (environment)	Health promoting behaviour—physical activity participation	CASP Qualitative Checklist Critical Appraisal Skills Programme (2018b) and the Community Guide to Preventative Services Data Abstraction Form (Part III) Zaza et al. (2000)	Built environment barriers identified were physical (e.g. safety, inadequate lighting, uneven sidewalks, barriers on paths), temporal (e.g. maintenance, pedestrian crossing timing) and behaviour of other people (e.g. crowded sidewalks).
Policy (financial) Physical (facility/equipment) Attitudes (providers) Services (transport) Knowledge/training (providers)	Access to healthcare	Rating method from the UK National Service Framework for Long-term Conditions Department of Health Long-Term Conditions NSF Team (2005)	Found that access to health services was limited by impairment and activity limitations (e.g. mobility, cognition, communication), contextual factors (e.g. cost, transport, physical barriers, clinician's attitudes, personal factors, such as age, socioeconomic status, caregiver health).
Physical (facility/equipment) Attitudes and support (providers) Knowledge/training (providers)	Access to healthcare; Care quality—maternal and infant health	CASP Qualitative Checklist Critical Appraisal Skills Programme (2018b)	Barriers identified included accessibility of facilities, lack of adapted equipment, lack of healthcare provider knowledge of physical disability and pregnancy, healthcare provider's dismissals of concerns and unwillingness to assist, disabled women's feelings of being dehumanised, afraid and vulnerable.
Policy (organisation) Attitudes and support (providers and community) Services (information provision/communication) Knowledge/training (providers)	Access to healthcare; Care quality—maternal and infant health	Quality Appraisal Framework Caldwell et al. (2011)	Four broad themes were identified 'in the family way' (i.e. physical changes, suitability as a mother), knowledge and advocacy (i.e. communication, decision-making), midwives educational needs (i.e. lack of knowledge of women with intellectual disability and pregnancy) and midwives attitudes (i.e. ability to manage taking care of child).
Physical (facility/equipment) Attitudes (providers) Services (information provision/communication)	Access to healthcare—accessibility of health information	Risk of bias for observational studies with Newcastle-Ottawa scale Wells et al. (2000), CASP Qualitative Checklist Critical Appraisal Skills Programme (2018b) and Quality Assessment Tool for Quantitative studies Effective Public Health Practice Project (2018)	Barriers to contraceptive knowledge, services or informed decision-making were limited access to plain-language information, limited knowledge and negative attitudes by health professionals, denied care because of disability, physically inaccessible clinics and exam tables and disabled person not involved in decision-making.
Policy (organisation) Physical (environment; facility/equipment) Attitudes (providers) Services (information provision/communication) Knowledge/training (providers)	Access to healthcare; Care quality—maternal and infant health	None specified	Barriers to service access included accessibility (e.g. physical location, non-flexible model of care, lack of transportation, physical access within facilities, lack of suitable equipment, fragmented and uncoordinated care), health professional's adaption to needs (e.g. lack of suitable health and parental education information) and acceptability (e.g. poor health professional knowledge of disability and negative attitudes). Facilitators to accessing services included improvements in the physical environment (e.g. facilities should adopt universal design principles, improving physical access to facilities, disabled parking, all amenities within reach), improvements in service provision (e.g. a disability advisor at the facility, services near residence (e.g. community-based), flexible care provision), information access (e.g. information in appropriate format that can be easily understood and is culturally sensitive), improve acceptability (e.g. disability awareness and training programs for health professionals, safe handling and transfer of women with a physical disability, early referral to allied health professionals for individual needs assessment).

(Continues)

TABLE 2 (Continued)

Authors	Review aim	Studies	Population	Disability
Malli et al. (2018)	Understand the impact of austerity policy on the lives of people with learning disabilities and effects on people who support them	11 studies: all qualitative	Children and adults, age unspecified	Intellectual or learning disability
Marrocco and Krouse (2017)	Understand the unique obstacles that persons with disabilities face in accessing preventive healthcare services	23 studies: 13 quantitative, 10 qualitative	Adults ≥18 years	Non-specific disability
Martin Ginis et al. (2016)	Identify barriers (and facilitators) to leisure-time physical activity participation among people with physical disabilities in review articles	22 studies: 13 systematic reviews, 9 other reviews (e.g. clinical guidelines, discussion or opinion pieces)	Children and adults, age unspecified	Neurological or physical disability
Matin et al. (2021)	Review barriers in access to healthcare services for women with disabilities	24 studies: all qualitative	Adults (women >18 years)	Physical, sensory and intellectual disability
Mulligan et al. (2012)	Assess environmental and personal barriers of physical activity participation in populations with long term neurological conditions	28 studies: 21 quantitative, 7 qualitative	Adults ≥18 years	Long-term neurological disability (≥6 months)
Newitt et al. (2016)	Describe factors that influence participation in physical activity for people with neuromusculoskeletal (NMS) conditions	15 studies: all qualitative	Adults ≥18 years	Physical disabilities

Environmental factor themes identified	Health outcome examined	Quality assessment tool used	Author conclusions
Policy (government; financial)	Care quality—personal care, activities of daily life	CASP Systematic Review Checklist Critical Appraisal Skills Programme (2018c)	Found that economic austerity measures were not aligned with supporting the care needs of people with learning disabilities and that financial cuts had adversely affected people with learning disabilities (e.g. social isolation) and their informal carers (e.g. increased demand on family carer's time, less labour market participation).
Policy (financial) Physical (environment; facility/equipment) Attitudes (providers) Services (transport) Knowledge/training (providers)	Access to healthcare—preventive	None specified	Barriers identified included the physical environment (e.g. inadequate ramps, parking, examination space), diagnostic equipment issues (e.g. suitability of mammogram equipment for wheelchair-users), transportation challenges, lack of provider knowledge about disability and poor care coordination, negative provider attitudes and financial issues.
Policy (government; organisation; financial) Physical (environment; facility/equipment) Attitudes and support (providers and community) Services (transport; available programs) Knowledge/training (providers)	Health-promoting behaviour—physical activity participation	A Measurement Tool to Assess Systematic Reviews (AMSTAR) Shea et al. (2009)	Barriers identified were intrapersonal (i.e. psychological e.g. negative attitudes and beliefs, body functions and structures e.g. pain, fatigue and employment status), interpersonal (i.e. lack of social support, negative attitudes of others and social pressures) and institutional (i.e. lack of disability-specific knowledge of people in facilities, poor rehabilitation processes, building design and construction and programme factors e.g. availability and enjoyable activities). Specific facilitators identified included positive attitudes and beliefs about being active and oneself.
Policy (financial) Physical (environment; facility/equipment) Attitudes and support (providers and family) Services (transport; information provision/communication) Knowledge/training (providers)	Access to healthcare	Consolidated Criteria for Reporting Qualitative Research O'Brien et al. (2014) and Tong et al. (2007)	Barriers identified included sociocultural (e.g. erroneous assumptions, negative attitudes, low health literacy), financial (e.g. unemployment, poverty, high transportation costs) and structural (e.g. lack of health insurance, inaccessible equipment, lack of knowledge of healthcare staff of disability, communication problems).
Physical (facility/equipment) Attitudes and support (providers and community) Services (transport; available programs; information provision/communication) Knowledge/training (providers)	Health-promoting behaviour—physical activity participation	Qualitative Appraisal Tool Checklist for interviews and focus groups Attree (2004) and Attree and Milton (2006) and Joanna Briggs Institute (JBI) Assessment for Descriptive Studies Joanna Briggs Institute (JBI) (2020)	Barriers included activity and participation (e.g. limited mobilisation, assistance with activities of daily living, embarrassment), physical environmental (e.g. available, accessible, affordable transport, cost of program or facility access, difficulty accessing assistance within facilities due to physical layout, lack of information about programs, lack of suitable programs, difficulty obtaining specialised exercise equipment), social environment (e.g. no expectation to be physically active, lack of knowledge of staff regarding requirements of disabled people to support participation), personal (e.g. age and unemployment).
Policy (organisation; financial) Physical (environment; facility/equipment) Attitudes and support (providers and community) Services (available programs; transport) Knowledge/training (providers)	Health promoting behaviour—physical activity participation	McMaster Critical Review Forms Law et al. (1998) and Letts et al. (2007)	Barriers identified were body functioning (e.g. fluctuating symptoms, fatigue, muscle weakness, pain, depression, fear of injury, balance, stability, communication difficulties), environmental (e.g. inaccessible buildings and pools, long travel distances to accessible facilities, no rest places or rails to hang onto near equipment, financial cost, lack of health professional knowledge of disability, negative societal attitudes), personal (e.g. lack of motivation, self-conscious, embarrassed, uncomfortable in new environment). Motivators to participate were categorised as body functioning (e.g. maintain independence and function, weight control), environmental (e.g. social support, safety devices present and assistance from health professionals), personal (e.g. goal setting and achieving, enjoyment, feelings of accomplishment).

TABLE 2 (Continued)

Authors	Review aim	Studies	Population	Disability
Trounson et al. (2020)	Identify factors that facilitate or impede the engagement of Aboriginal people living with disability with disability services	20 studies: 17 qualitative, 2 theoretical, 1 book	Children and adults (Aboriginal and Torres Strait Islander people), age unspecified	Non-specific disability
Wilbur et al. (2019)	Assess the menstrual hygiene requirements of disabled people, the barriers they face and available interventions	22 studies: 14 quantitative, 7 qualitative, 1 quasi-experimental	Specified as people who menstruate and/or their carers, age unspecified	Non-specific disability
Yang et al. (2020)	Identifying factors influencing self-care behaviours in adults with diabetes and disability	20 studies: 7 qualitative, 6 quantitative, 6 quasi-experiment, 1 participatory action research	Adults ≥18 years with diabetes	Non-specific disability

person's access to health services (Edwards et al., 2020; Gibson & O'Connor, 2010; Lawler et al., 2013; Marrocco & Krouse, 2017; Matin et al., 2021; Trounson et al., 2020) and other recreational facilities (Bodde & Seo, 2009; Martin Ginis et al., 2016; Mulligan et al., 2012; Newitt et al., 2016). Transport accessibility was not just a matter of physical accessibility, but also included the available routes, the reliability of the service and financial cost. The ways in which transport impacted access for disabled people was multifaceted, overlapping with barriers and enablers produced by the physical environment (e.g. whether public buses or trains could be accessed independently, distance from nearest transit stop to a facility), organisational and financial aspects (e.g. set appointment times, cost of private transport) and social attitudes (Edwards et al., 2020; Matin et al., 2021). Transportation policies and services should consider physically accessibility issues, along with other barriers (such as cost and service frequency), to avoid social exclusion of disabled people.

Within rural and regional areas, a lack of health professionals and specialist healthcare services, along with high turnover of staff, was identified in three reviews (Barr et al., 2018; Lawler et al., 2013; Matin et al., 2021) as an impediment to care provision and care quality. The absence of local services necessitated travelling long distances to access specialised healthcare which, in turn, resulted in increased financial burden due to costs associated with work absence, travel and accommodation (Barr et al., 2018). However, the

increasing availability of telemedicine services for disabled people was acknowledged as a potential resource to ameliorate the absence of local specialist care delivery (Barr et al., 2018).

4.4 | Information, communication and care coordination

Inaccessible information and communication barriers were found to impact both on disabled people's access to, and utilisation of, health-care, the quality of care received, and their ability to be fully informed decision-makers in their own care (Beverley et al., 2004; Bradbury-Jones et al., 2013; Edwards et al., 2020; Heideveld-Gerritsen et al., 2021; Horner-Johnson et al., 2019; Yang et al., 2020). Common barriers included the provision of inaccessible or confusing information in letters, referrals, brochures and medication labels (Beverley et al., 2004; Homeyard et al., 2016; Matin et al., 2021). The commonplace use of jargon and medical terminology was especially seen as a barrier for people with cognitive or intellectual impairments (Bradbury-Jones et al., 2013). Within the acute care setting, a limited variety of visual or auditory media were used to convey health information. Where used, resources such as plain-English factsheets, large print text, tactile maps, augmentative and alternative communication, and audio material enabled disabled people to exert their

Environmental factor themes identified	Health outcome examined	Quality assessment tool used	Author conclusions
Policy (government; organisation; financial) Attitudes (providers) Services (transport; information provision/ communication)	Care quality; Access to healthcare	CASP Qualitative Checklist Critical Appraisal Skills Programme (2018b)	Barriers included culture and Indigeneity (e.g. providers need to be more aware of impact of Indigenous culture, disability as a construct not a defining term, difficulty performing some cultural activities, need for strong community and family support network), accessibility (e.g. difficulties engaging with medical or government services, historical disablement, inflexible access pathways, complex language), engagement (e.g. service design does not account for cultural disparities, poor communication between service providers and consumers, lack of Indigenous inclusion and consultation in service design), lack of support (e.g. economic, isolation, psychological support needs not met).
Attitudes and support (providers and family) Services (personal equipment; information provision/ communication)	Health-promoting behaviour—menstrual hygiene	Modified versions of STROBE and RATS Von Elm et al. (2007), risk of bias assessed using method by Banks et al. (2017)	Barriers included a lack of standard guidance for carers, lack of menstruation training, information and support for people with intellectual disabilities and their carers, cost of menstrual products, lack of appropriate alternative options for people with physical impairments.
Policy (financial) Physical (environment) Attitudes and support (providers and family) Services (personal equipment; information provision/ communication) Knowledge/training (providers)	Health promoting behaviour—diabetes self-care	JBIC Critical Appraisal Checklists Joanna Briggs Institute (JBIC) (2020)	Barriers included other comorbidities, negative thoughts. Barriers specific to people with physical disabilities were mobility limitations, medically underserved areas and feeling overwhelmed. Barriers specific to people with visual impairments were a lack of accessible information on diabetes. Facilitators included education, motivation and confidence. Support for people with developmental disabilities and helpful assistive devices and modifications to physical environments for people with visual impairments.

right to informed consent, communication and choice (Beverley et al., 2004; Bradbury-Jones et al., 2013; Homeyard et al., 2016) and are recommended.

The reviews identified a lack of clear information available to disabled people and their families in terms of available services and how to navigate them. Two reviews highlighted the deficiency of tailored health advice and information for individuals with disabilities, for example guidance on undertaking physical activity with a disability (Mulligan et al., 2012; Newitt et al., 2016). Reviews focussing on disabled women's experiences reported the limited availability of relevant, high-quality information on contraception, pregnancy, post-partum support and parenting (Heideveld-Gerritsen et al., 2021; Horner-Johnson et al., 2019; Lawler et al., 2013; Matin et al., 2021). Even when credible and relevant information was available on the internet, for example, professional assistance may still be required to assist in locating and understanding the health messaging (Barr et al., 2018). Complex language and complicated service structures were identified as leading to confusion and delayed access to services (Beverley et al., 2004; Trounson et al., 2020). There was evidence in the reviews that poor communication between providers, or between providers and disabled people and their families, resulted in poor care coordination between services and fragmented care (Barr et al., 2018; Lawler et al., 2013). Improving care coordination could involve strategies, such as developing health information

for disabled people that is disability-specific and encouraging care providers to participating in communication skills training.

4.5 | Government, organisational and financial policy

The wide-ranging impact of government policies that determine the provision of health and social services, allocate funding for programs, set transport policies and legislate and enforce accessibility standards on the lives of disabled people was explicitly acknowledged in some reviews (Gibson & O'Connor, 2010; Marrocco & Krouse, 2017; Martin Ginis et al., 2016; Trounson et al., 2020:14). One review specifically examined the impact of government austerity measures in the United Kingdom on the quality of care for disabled people and found a lowering of the quality of care and support provided to a disabled person to achieve ADLs, leading to an increase in crisis health interventions rather than regular participation in preventive healthcare practices (Malli et al., 2018).

Organisational policies were viewed as creating barriers to service and care access in the form of eligibility criteria and inflexible models of care. In Australia, rigid intake streams and eligibility criteria for services can exclude Aboriginal people who have not obtained sufficient diagnostic testing due to limited

TABLE 3 Quality assessment using the JBI Critical Appraisal Checklist for Systematic Reviews and Research Syntheses^b

Authors	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Quality ^a
Barr et al. (2018)	Y	Y	Y	Y	Y	Y	N	Y	NA	Y	Y	High
Beverley et al. (2004)	Y	Y	Y	Y	Y	Y	N	Y	Y	Y	Y	High
Bloemen et al. (2015)	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	High
Bodde and Seo (2009)	Y	Y	Y	Y	Y	U	N	Y	U	Y	Y	High
Bradbury-Jones et al. (2013)	Y	Y	Y	Y	U	U	U	Y	N	Y	Y	High
Breckenridge et al. (2014)	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	High
Da Rosa et al. (2020)	Y	U	Y	Y	Y	Y	Y	Y	Y	Y	Y	Mid
Edwards et al. (2020)	Y	Y	Y	Y	Y	Y	Y	Y	N	Y	Y	High
Eisenberg et al. (2017)	Y	Y	Y	Y	Y	Y	Y	Y	U	Y	Y	High
Gibson and O'Connor (2010)	Y	U	U	Y	Y	U	N	Y	N	Y	Y	Mid
Heideveld-Gerritsen et al. (2021)	Y	Y	Y	Y	Y	Y	U	Y	U	Y	Y	Mid
Homeyard et al. (2016)	Y	U	Y	Y	Y	Y	Y	Y	Y	Y	Y	High
Horner-Johnson et al. (2019)	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	High
Lawler et al. (2013)	Y	Y	Y	Y	U	U	U	U	U	Y	Y	Low
Malli et al. (2018)	Y	Y	Y	Y	Y	Y	Y	Y	N	Y	NA	High
Marrocco and Krouse (2017)	Y	Y	Y	Y	N	N	U	U	N	Y	Y	Mid
Martin Ginis et al. (2016)	Y	Y	Y	Y	Y	Y	Y	Y	N	Y	NA	High
Matin et al. (2021)	Y	Y	Y	Y	Y	Y	Y	Y	N	Y	Y	High
Mulligan et al. (2012)	Y	Y	Y	Y	Y	Y	Y	Y	N	Y	NA	High
Newitt et al. (2016)	Y	Y	Y	Y	Y	U	Y	Y	NA	U	N	Mid
Trounson et al. (2020)	Y	Y	Y	Y	Y	U	Y	Y	NA	Y	Y	High
Wilbur et al. (2019)	Y	Y	Y	Y	Y	U	Y	Y	NA	NA	Y	Mid
Yang et al. (2020)	Y	Y	Y	Y	Y	Y	Y	Y	NA	Y	Y	High

Abbreviations: N, No; NA, Not applicable; U, Unclear; Y, Yes.

^aJoanna Brigg's Institute (JBI) critical appraisal checklist for systematic reviews and research syntheses, checklist items marked 'yes': <5 = low quality; 6–8 = mid quality; >9 = high quality.

^bJBI Critical Appraisal Checklist questions:

1. Is the review question clearly and explicitly stated?
2. Were the inclusion criteria appropriate for the review question?
3. Was the search strategy appropriate?
4. Were the sources and resources used to search for studies adequate?
5. Were the criteria for appraising studies appropriate?
6. Was critical appraisal conducted by two or more reviewers independently?
7. Were there methods to minimise errors in data extraction?
8. Were the methods used to combine studies appropriate?
9. Was the likelihood of publication bias assessed?
10. Were recommendations for policy and/or practice supported by the reported data?
11. Were the specific directives for new research appropriate?

prior interaction with health services (Trounson et al., 2020:14). Appointment inflexibility (e.g. long waits, set appointment times, short consultations) was another frequently identified barrier in the reviews impacting on the use of healthcare facilities, including antenatal, hospital and secondary healthcare services (Homeyard et al., 2016; Lawler et al., 2013). Government agencies and organisations need to consider the health and service needs of disabled

people within their health- and social-related policies, especially service availability.

While lack of financial resources is an often-cited barrier for the general population, it is especially prohibitive for people with disabilities who often experience financial constraints and are economically vulnerable (Bodde & Seo, 2009; Breckenridge et al., 2014; Marrocco & Krouse, 2017; Matin et al., 2021). Two reviews identified

TABLE 4 Environmental factor theme supporting quotations

CHIEF theme	Sub-theme	Quotation	Authors
Policies	Policy—government	"[themes emerging at the policy level]...frequently cited factors within the potential purview of government policy-makers: health policies (specifically funding for programmes) and transportation services, systems, and policies." (p487)	Martin Ginis et al. (2016)
		"In terms of policy, the use of language, and the understanding of the cultural nuances about language, cultural understanding of disability and conditions associated with disability is exceptionally important when considering designing and delivering support services." (p21)	Trounson et al. (2020)
	Policy—financial	"Cost is likely to be even more prohibitive for adults with ID [intellectual disability], many of whom live on very limited or fixed resources." (p63)	Bodde and Seo (2009)
		"..report the cost of care and medication being an issue for disabled people receiving or being satisfied with medical services." (p26)	Gibson and O'Connor (2010)
	Policy—organisation	"Policies and guidelines that promote [physical] access should be devised, implemented, and audited on a regular basis." (p213)	Lawler et al. (2013)
		"..there is still considerable work to be done in the area of accessibility with most organisations addressing provision for individuals with disabilities as policy add-on rather than forming an integral part of sports development strategies." (p5)	Newitt et al. (2016)
Physical or structural	Physical—environment	"Physical accessibility to the building is the first and most obvious challenge encountered by women with a physical disability when accessing maternity services." (p211)	Lawler et al. (2013)
		"Several studies identified physical obstacles as one reason for not accessing care and ultimately not receiving care." (p285)	Marrocco and Krouse (2017)
	Physical—facility/equipment	"Physical inaccessibility [i.e. improper examination tables and inadequate equipment] is a major barrier to the effective utilisation of maternity services." (p5)	Breckenridge et al. (2014)
		"Disabled patients find it difficult to assess physicians' offices and, once inside, experience other barriers such as fixed height examination couches that impede transfers." (p26)	Gibson and O'Connor (2010)
Services or assistance	Services—available programs/services	"..families living in RRR [regional, rural and remote] areas had difficulties accessing specialist medical care and experienced allied health professional." (p121)	Barr et al. (2018)
		"Participants...travelled long distances to participate in exercise due to a lack of nearby accessible facilities." (p5)	Newitt et al. (2016)
	Services—information provision/communication	"Families of children with a disability reported a lack of information in RRR [regional, rural and remote] areas." (p121)	Barr et al. (2018)
		"..the latest information regarding diabetes was not readily shared in accessible ways and that diabetes educational forums attended by visually impaired people often lacked accessible non-visual technology." (p11)	Yang et al. (2020)
	Services—transport	"A consequence of the reduced choice of services in RRR [regional, rural and remote] areas is extensive travel for families to larger centres and cities." (p121)	Barr et al. (2018)
		"Organising transport to appointments was a concern along with unreliability, long waits, dealing with rude drivers, getting into and out of the vehicles, and having to cancel at short notice, along with having to take public transport which could often exacerbate existing symptoms." (p7)	Edwards et al. (2020)

(Continues)

TABLE 4 (Continued)

CHIEF theme	Sub-theme	Quotation	Authors
	Services—personal equipment	"...people with spinal cord injuries...reported discomfort and difficulty in positioning the menstrual product to ensure its maximum absorbency, as well as increasing difficulties with catheters and urinary management during menstruation." (p10)	Wilbur et al. (2019)
		"Intervention to enhance blood glucose self-monitoring using other senses, including touch and hearing, were helpful in maintaining appropriate glucose levels or in increasing the quality of blood glucose monitoring." (p12)	Yang et al. (2020)
Attitude or support	Attitudes and support—providers, community, family	"Strong social support from family, friends, and fitness staff is actually one of the best predictors of participating in regular PA [physical activity]." (p62)	Bodde and Seo (2009)
		"There is strong evidence that women with physical disabilities encounter unwillingness to provide assistance on the part of healthcare providers." (p9)	Heideveld-Gerritsen et al. (2021)
	Knowledge/training—providers	"Lack of basic understanding reduced nurses' ability to pick up on cues, and they recognised the impact of this on their competence and confidence in caring." (p1504)	Bradbury-Jones et al. (2013)
		"While specialists may lack knowledge about general health issues, general practitioners may lack knowledge about outcomes in specific disabling conditions." (p26)	Gibson and O'Connor (2010)

Abbreviation: CHIEF, Craig Hospital Inventory of Environmental Factors.

that some health insurance companies in the United States did not cover the additional healthcare needs (e.g. additional examination time or procedures) of disabled people (Breckenridge et al., 2014; da Rosa et al., 2020), compounding the financial burden on the disabled person. In some instances, reviews also reported that disabled people did not obtain health insurance because of the financial cost, and the limited coverage of services that they required (Marrocco & Krouse, 2017; Matin et al., 2021). Likewise, the cost of health-promoting behaviours that cater to people with disabilities, such as physical activity programs, may also prohibit participation (Bodde & Seo, 2009; Mulligan et al., 2012).

4.6 | Attitudes and social support

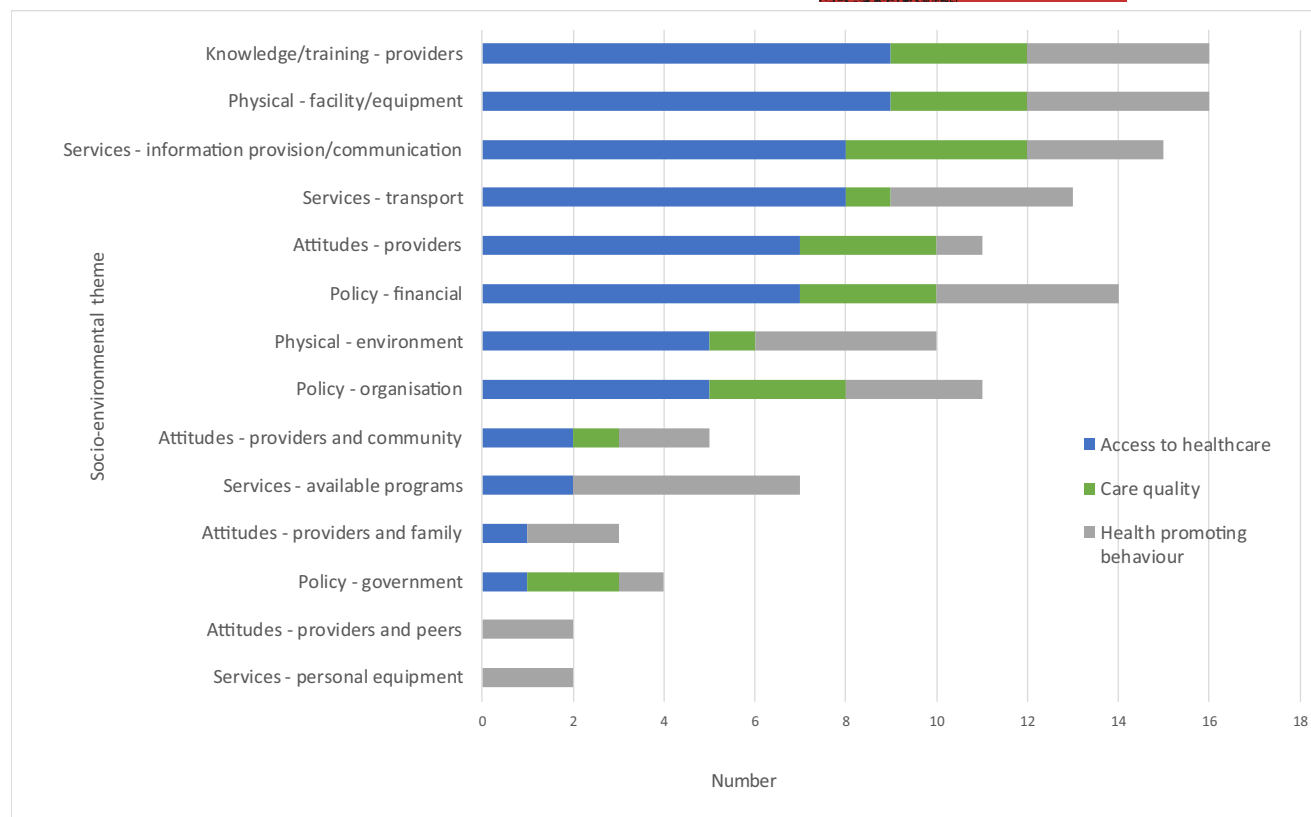
Several reviews touched on the significance of the social environment in encouraging people with physical disabilities to participate in health-promoting behaviours like physical activity (Bloemen et al., 2015; Bodde & Seo, 2009; Martin Ginis et al., 2016; Mulligan et al., 2012; Newitt et al., 2016). Reviews reported that disabled people can feel embarrassed to exercise in public, be wary of stereotyping and negative perceptions of disability, and become demotivated by the low expectations and lack of encouragement to engage from staff, family and peers (Bloemen et al., 2015; Martin Ginis et al., 2016; Mulligan et al., 2012; Newitt et al., 2016). However, evidence from the reviews also supported the converse. Participation in physical activity among disabled people was facilitated by positive support from family and the community, information on how to be involved, and encouragement and advice from health and exercise

professionals (Bloemen et al., 2015; Martin Ginis et al., 2016; Newitt et al., 2016).

In relation to the utilisation of health services, it was also noted that '*improving access to [...] care cannot be achieved by addressing internal service barriers alone*' (Breckenridge et al., 2014, p.9). Reviews suggested that investing in community engagement and providing opportunities for social interaction extending to people with and without disabilities would likely strengthen support, increase engagement with services and improve health and wellbeing for disabled people (Breckenridge et al., 2014; Trounson et al., 2020). In particular, it was acknowledged that adults with learning disabilities should be supported to make decisions regarding their own care based on their unique, individual needs and preferences (Bradbury-Jones et al., 2013; Horner-Johnson et al., 2019; Matin et al., 2021). Moreover, adults with learning disabilities should also be involved with policy development, awareness raising, service evaluation and research (Bradbury-Jones et al., 2013).

4.7 | Implications for policy, practice and research

There were common and cross-cutting themes related to socio-environmental themes identified across different disabilities, age groups and types of health outcomes in the reviews. These commonalities indicate that further work needs to be done to ensure services are inclusive, health professionals are adequately trained and knowledgeable about providing disability-focused care, and universal design principles are adopted for the built environment and medical equipment. Government policies should consider transportation



¹Multiple themes and health outcomes could be identified for each systematic review.

FIGURE 2 Number of socio-environmental themes identified by health outcome for disabled people¹

issues and service availability in regional areas, including telehealth capacity and availability. Plain-language messaging in different formats (e.g. large print, tactile) and involving disabled people in service design is vital.

There were certain population sub-groups where there was a lack of reviews examining social, physical or attitudinal factors influencing health outcomes, including for disabled children (i.e. ≤ 16 years) and older people (i.e. ≥ 65 years). With the ageing of the population in many high-income countries (United Nations Department of Economic & Social Affairs: Population Division, 2020), there are likely to be key factors relating to ageing and disability that are unacknowledged. Reviews focusing solely on disabled men were largely absent, particularly for preventive health behaviour, such as cancer screening services (Edwards et al., 2020).

Overall, there was limited information on specific health outcomes (e.g. care complications, hospital readmissions) of disabled people using primary, acute, sub-acute or community-based health services. This is predominantly due to disability not being recorded or easily identified in routine administrative data collections, such as hospitalisation, emergency department or primary care data (Halcomb et al., 2019). There are increasing opportunities for better identification of disabled people in health data collections through linking health and social data collections. Assessing publication bias in systematic reviews aids in limiting bias towards the inclusion of studies in reviews based on the

strength or direction of supporting evidence (Ayorinde et al., 2020). That only six of the reviews in the current overview assessed the risk of publication bias opens some reviews up to selection bias, which could influence review outcomes.

4.8 | Strengths and limitations

The strengths of this overview of systematic reviews were the comprehensive search strategy, that the review followed the PRISMA guideline, and multiple reviewers were involved in the data extraction and synthesis phases with high interrater reliability. Any disagreements were discussed between reviewers with consensus achieved. The overview also had several limitations. Only systematic reviews were considered, therefore other reviews (such as integrative and scoping reviews) were excluded that may have contributed additional insights. There is a potential risk of bias in the identification of the key socio-environmental factors due to the possible overrepresentation of individual studies that were included in multiple systematic reviews. Reviews published in non-English languages and that considered studies only from low- and middle-income countries were excluded, which may result in language- and generalisability-bias respectively.

5 | CONCLUSIONS

Disabled people encounter common social, physical and attitudinal factors that can hinder their health outcomes in terms of access to services and quality healthcare. Many preventive health and health-care services were identified as either inaccessible or not meeting the needs of disabled people. There is an obvious need to reduce health and social inequalities in access to healthcare and participation for disabled people. Greater involvement of disabled people in service design and awareness raising is essential.

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AUTHOR CONTRIBUTIONS

R. Mitchell: Conceptualisation, Methodology, Investigation; Writing-Original draft preparation, Writing-Review & Editing; T Ryder: Methodology, Investigation, Writing-Original draft preparation, Writing-Review & Editing; K. Matar: Investigation; R. Lystad: Writing-Review & Editing; R. Clay-Williams: Writing-Review & Editing; J. Braithwaite: Writing-Review & Editing.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no datasets were generated or analysed during the current study.

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